Lab 8: Raft Consensus Algorithm using Docker

1. Aim

To implement a three-node distributed system using the Raft consensus algorithm inside Docker containers and observe leader election, follower nodes, and fault tolerance.

2. Learning Objectives

After completing this experiment students will be able to:

- Understand distributed systems
- Understand leader election in Raft
- Use Docker to simulate distributed machines
- Observe fault tolerance when a leader fails
- Understand container networking

3. Software Requirements

Students must install the following software:
Docker Desktop Python (inside container) Command Prompt / Terminal
To verify Docker installation run:
docker --version

4. Create Project Folder

Step 1
Open File Explorer
Go to Desktop
Create a new folder named:
raft_docker_lab

5. Open Terminal

Open Command Prompt
Navigate to the project folder
cd %USERPROFILE%\\OneDrive\\Desktop\\raft_docker_lab

6. Create Required Files

Students must create four files inside the folder.
Files:
node.py Dockerfile requirements.txt docker-compose.yml

Inside the terminal run this

type nul > node.py & type nul > Dockerfile & type nul > requirements.txt & type nul > docker-compose.yml

Then check:

dir

Folder Structure

```
raft_docker_lab
│
├── node.py
├── Dockerfile
├── docker-compose.yml
└── requirements.txt
```

Take Screenshot 1: Folder structure

```
C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>dir
 Volume in drive C is Windows
 Volume Serial Number is F23A-D83D

 Directory of C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab

16-03-2026  22:09    <DIR>          .
16-03-2026  22:07    <DIR>          ..
16-03-2026  22:09                 0 docker-compose.yml
16-03-2026  22:09                 0 Dockerfile
16-03-2026  22:09                 0 node.py
16-03-2026  22:09                 0 requirements.txt
               4 File(s)              0 bytes
               2 Dir(s)  343,472,115,712 bytes free

C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>
```

7. Create requirements.txt

Command: **notepad requirements.txt**

Paste the following

raftos

Save as
requirements.txt

8. Create Dockerfile

Command**: notepad Dockerfile**
Write the code
Save as:
Dockerfile

9. Create docker-compose.yml

Command**: notepad docker-compose.yml**
Write the code

10. Create node.py

Command: **notepad node.py**
Write the code
Save as: **node.py**

11. Build and Run Containers

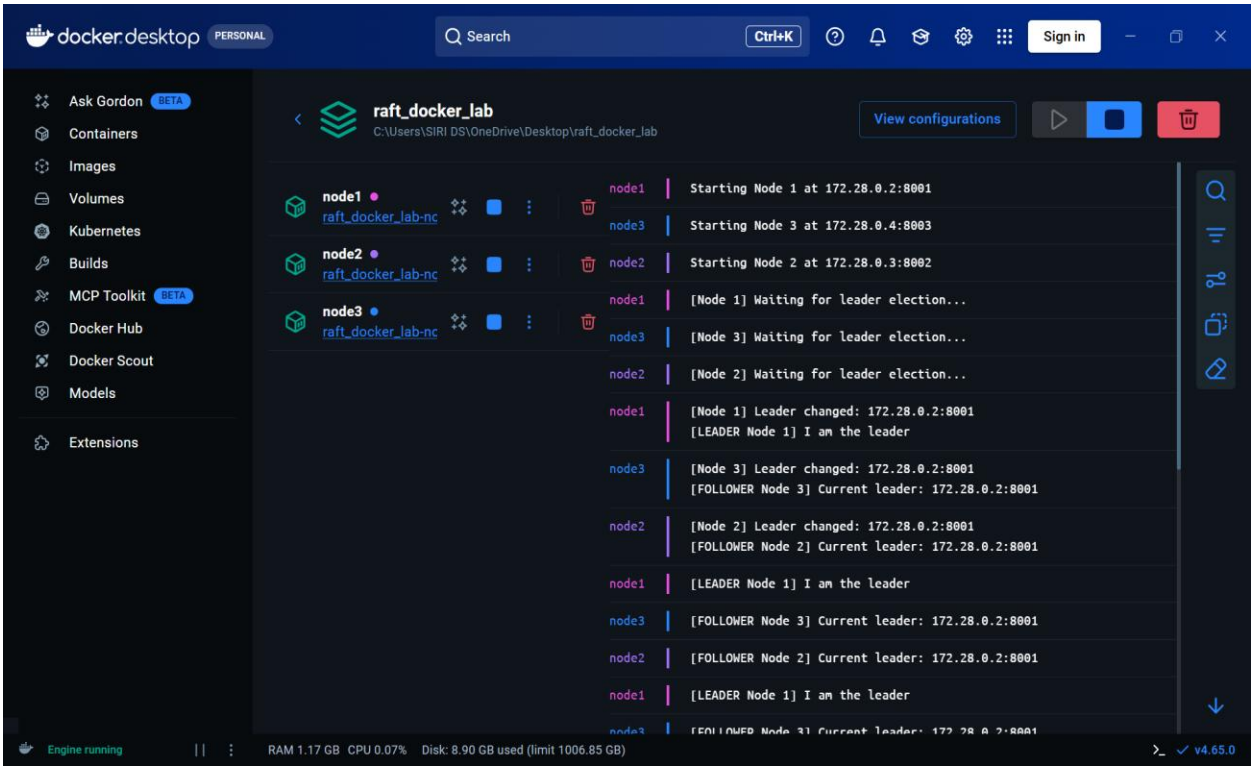
Open terminal
Run:
docker compose up -—build

Take Screenshot 2A: Docker build process (Terminal)

```
#20 [node3] resolving provenance for metadata file
#20 DONE 0.0s
[+] up 7/7
 ✔Image raft_docker_lab-node2    Built                                    4.7s
 ✔Image raft_docker_lab-node3    Built                                    4.7s
 ✔Image raft_docker_lab-node1    Built                                    4.7s
 ✔Network raft_docker_lab_raftnet Created                                 0.1s
 ✔Container node3                Started                                  0.8s
 ✔Container node2                Started                                  0.8s
 ✔Container node1                Started                                  0.7s

C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>
```

Take Screenshot 2B: Docker build process (Docker)



12. Observe Leader Election

Observe Leader Election
After starting the containers, students must observe the logs of each node to understand how the Raft algorithm elects a leader.

Step 1: Open Three Separate Terminals
Open three new Command Prompt windows (Terminal 2, Terminal 3, and Terminal 4).
Navigate to the project folder in each terminal:
cd %USERPROFILE%\OneDrive\Desktop\raft_docker_lab
Step 2: Observe Logs of Each Node
Run the following command in Terminal 2:
docker logs -f node1
Run the following command in Terminal 3:
docker logs -f node2

Run the following command in Terminal 4:

docker logs -f node3

The -f option allows the terminal to continuously display live logs from the container.

Step 3: Observe Leader Election

Students should observe the outputs appearing in the terminals. One node will become the Leader, while the other nodes will act as Followers.

Step 4: Observe Logs in Docker Desktop

Students can also observe the same logs through the Docker Desktop interface by selecting the running containers node1, node2, and node3 and viewing their logs

Take Screenshot 3A: Leader election logs from docker

```
node1   [LEADER Node 1] I am the leader

node2   [FOLLOWER Node 2] Current leader: 172.28.0.2:8001

node3   [FOLLOWER Node 3] Current leader: 172.28.0.2:8001
```

Take Screenshot 3B: Leader election logs from Terminal 1

```
Command Prompt    Command Prompt - docker l    Command Prompt - docker lo
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\SIRI DS>cd %USERPROFILE%\\OneDrive\\Desktop\\raft_docker_lab

C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>docker logs -f node1
Starting Node 1 at 172.28.0.2:8001
[Node 1] Waiting for leader election...
[Node 1] Leader changed: 172.28.0.2:8001
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
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[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
```

Take Screenshot 3C: Leader election logs from Terminal 2

```
Command Prompt    Command Prompt - docker lo    Command Prompt - docker l

Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\SIRI DS>cd %USERPROFILE%\\OneDrive\\Desktop\\raft_docker_lab

C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>docker logs -f node2
Starting Node 2 at 172.28.0.3:8002
[Node 2] Waiting for leader election...
[Node 2] Leader changed: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
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[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
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[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
```

Take Screenshot 3D: Leader election logs from Terminal 3

```
Command Prompt    Command Prompt - docker lo    Command Prompt - docker lo

(c) Microsoft Corporation. All rights reserved.

C:\Users\SIRI DS>cd %USERPROFILE%\\OneDrive\\Desktop\\raft_docker_lab

C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>docker logs -f node3
Starting Node 3 at 172.28.0.4:8003
[Node 3] Waiting for leader election...
[Node 3] Leader changed: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
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[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
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[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
```

13. Check Running Containers

Open another terminal

Run:

docker ps

Take Screenshot 4: Running containers

```
Command Prompt   X   Command Prompt - do(   X   Command Prompt - do(   X   Command Prompt - do(   X   Command Prompt   X

Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\SIRI DS>cd %USERPROFILE%\\OneDrive\\Desktop\\raft_docker_lab

C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>docker ps
CONTAINER ID   IMAGE                  COMMAND              CREATED        STATUS         PORTS     NAMES
50862191db15   raft_docker_lab-node1  "python node.py 1"   3 minutes ago  Up 3 minutes             node1
5791bad1b308   raft_docker_lab-node2  "python node.py 2"   3 minutes ago  Up 3 minutes             node2
ebc394b8b885   raft_docker_lab-node3  "python node.py 3"   3 minutes ago  Up 3 minutes             node3

C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>
```

14. Check Docker Network

Run:

docker network inspect raft_docker_lab_raftnet

Take Screenshot 5: Docker network configuration

Command Prompt | Command Prompt - doc | Command Prompt - doc | Con

```
✔Container node1                Started

C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>docker network inspect raft_docker_lab_raftnet
[
    {
        "Name": "raft_docker_lab_raftnet",
        "Id": "f1222a34cd89fa04f4bb28478d5d93e39085bedc01ebeeedf70b2864b340985d",
        "Created": "2026-03-16T17:50:54.92932956Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "172.28.0.0/16",
                    "Gateway": "172.28.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.enable_ipv4": "true",
            "com.docker.network.enable_ipv6": "false"
        },
        "Labels": {
            "com.docker.compose.config-hash": "7b955128463677559d4edba13e9fe8f69dd803b2dfc190dd0f2333946fc6a5d0",
            "com.docker.compose.network": "raftnet",
            "com.docker.compose.project": "raft_docker_lab",
            "com.docker.compose.version": "5.1.0"
        },
        "Containers": {
            "50862191db154eb1c32cc7f17511643e167457a365c9df24a2b5ef5b558273d2": {
                "Name": "node1",
                "EndpointID": "0654358db2558fe5f71c6e459a2085403aaa87bdeadd42ba8501c0f0503e8097",
                "MacAddress": "ee:1c:f5:de:ce:ec",
                "IPv4Address": "172.28.0.2/16",
                "IPv6Address": ""
            },
            "5791bad1b308b1ba6692f6337de6fb3c9184a1f9301437711979be93b77a4cf3": {
                "Name": "node2",
                "EndpointID": "1b1ac8ccb0d29ab874f0d53e65641d2b545b58009923a7c36bf60a95c15dd884",
                "MacAddress": "e6:21:3f:a0:c7:b1",
                "IPv4Address": "172.28.0.3/16",
                "IPv6Address": ""
            },
            "ebc394b8b88586e51a4f5bf1114633bd201609faacf6294f5dc8f19d8735358f": {
                "Name": "node3",
                "EndpointID": "af8ff21c10642ada019d7d6840570e228f229cec6cb15685a5b414d5ef140666",
                "MacAddress": "ba:19:64:88:a7:5a",
                "IPv4Address": "172.28.0.4/16",
                "IPv6Address": ""
            }
        },
        "Status": {
            "IPAM": {
                "Subnets": {
                    "172.28.0.0/16": {
```

15. Simulate Leader Failure

Run:

docker stop node1

System will elect a new leader.

Take Screenshot 6A: Leader re-election

```
node2   [Node 2] Leader changed: None
        [Node 2] Waiting for leader election...

node3   [Node 3] Leader changed: None
        [Node 3] Waiting for leader election...

node2   [Node 2] Leader changed: 172.28.0.4:8003
        [FOLLOWER Node 2] Current leader: 172.28.0.4:8003

node3   [Node 3] Leader changed: 172.28.0.4:8003
        [LEADER Node 3] I am the leader

node2   [FOLLOWER Node 2] Current leader: 172.28.0.4:8003

node3   [LEADER Node 3] I am the leader

node2   [FOLLOWER Node 2] Current leader: 172.28.0.4:8003

node3   [LEADER Node 3] I am the leader
```

Take Screenshot 6B: Leader election logs from Terminal 2

```
[Node 2] Leader changed: None
[Node 2] Waiting for leader election...
[Node 2] Leader changed: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
```

Take Screenshot 6C: Leader election logs from Terminal 3

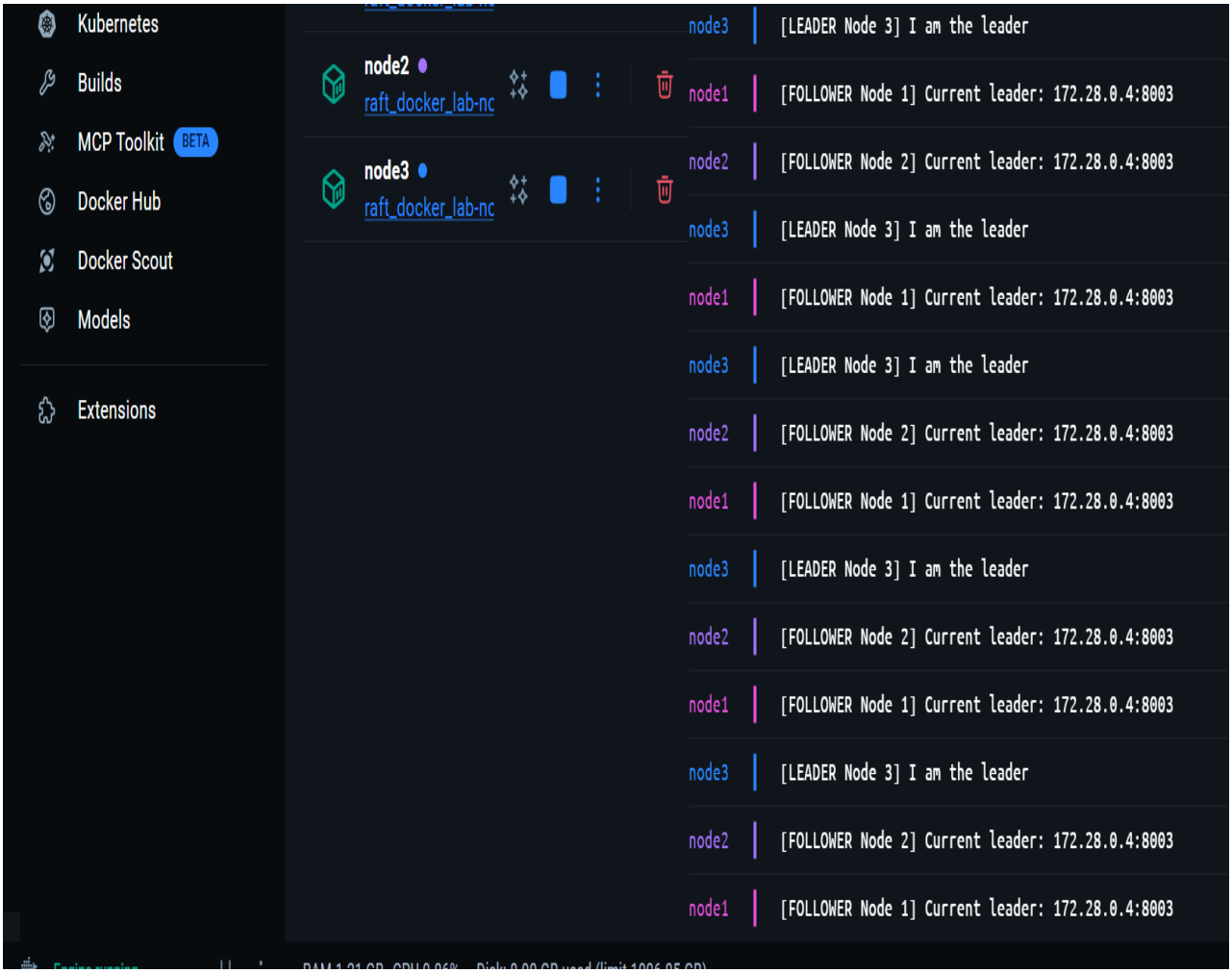
```
[Node 3] Leader changed: None
[Node 3] Waiting for leader election...
[Node 3] Leader changed: 172.28.0.4:8003
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
```

16. Restart Node

docker start node1

Node rejoins cluster as follower.

Take Screenshot 7A: Node Rejoining Cluster as Follower from docker



Run: docker logs -f node1

Take screenshot 7B: Node Rejoining Cluster as Follower from Terminal 1

```
[LEADER Node 1] I am the leader
Starting Node 1 at 172.28.0.2:8001
[Node 1] Waiting for leader election...
[Node 1] Leader changed: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
```

17. Stop Cluster

docker compose down

Activity:

Students must modify the existing 3-node setup to create a 5-node cluster on their own by updating the configuration files (docker-compose.yml and node.py). After successfully setting up the 5-node cluster, perform the following steps to observe quorum (majority) behavior in Raft.

18. Majority / Quorum Experiment using 5-Node Cluster

Open another terminal

Run:

docker compose up --build

Wait until all 5 containers start.

Check Running Containers

Open another terminal

Run:

docker ps

Take Screenshot 8: Running 5-node cluster

```
Microsoft Windows [Version 10.0.26200.8037]
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C:\Users\SIRI DS>cd %USERPROFILE%\\OneDrive\\Desktop\\raft_docker_lab

C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>docker ps
CONTAINER ID   IMAGE                  COMMAND              CREATED          STATUS          PORTS     NAMES
d926d88849cb   raft_docker_lab-node4  "python node.py 4"   47 seconds ago   Up 46 seconds             node4
a2b8fe9fd4f9   raft_docker_lab-node5  "python node.py 5"   47 seconds ago   Up 46 seconds             node5
d61948e3960f   raft_docker_lab-node1  "python node.py 1"   47 seconds ago   Up 46 seconds             node1
f7ece4e2cdd2   raft_docker_lab-node2  "python node.py 2"   47 seconds ago   Up 46 seconds             node2
3903be260355   raft_docker_lab-node3  "python node.py 3"   47 seconds ago   Up 45 seconds             node3

C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>
```

Observe Leader Election

Open 5 different terminals

Run:

docker logs -f node1

Take Screenshot 9:

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\SIRI DS>cd %USERPROFILE%\\OneDrive\\Desktop\\raft_docker_lab

C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>docker logs -f node1
Starting Node 1 at 172.28.0.2:8001
[Node 1] No leader elected yet
[Node 1] Leader changed: 172.28.0.2:8001
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
```

Run: docker logs -f node2

Take Screenshot 10:

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\SIRI DS>cd %USERPROFILE%\\OneDrive\\Desktop\\raft_docker_lab

C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>docker logs -f node2
Starting Node 2 at 172.28.0.3:8002
[Node 2] No leader elected yet
[Node 2] Leader changed: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
```

Run:docker logs -f node3

Take Screenshot 11:

```
C:\Users\SIRI DS>cd %USERPROFILE%\\OneDrive\\Desktop\\raft_docker_lab

C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>docker logs -f node3
Starting Node 3 at 172.28.0.4:8003
[Node 3] No leader elected yet
[Node 3] Leader changed: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
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[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
```

Run:docker logs -f node4

Take Screenshot 12:

```
C:\Users\SIRI DS>cd %USERPROFILE%\\OneDrive\\Desktop\\raft_docker_lab

C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>docker logs -f node4
Starting Node 4 at 172.28.0.5:8004
[Node 4] No leader elected yet
[Node 4] Leader changed: 172.28.0.2:8001
[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
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[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
```

Run:docker logs -f node5

Take Screenshot 13:

```
C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>docker logs -f node5
Starting Node 5 at 172.28.0.6:8005
[Node 5] No leader elected yet
[Node 5] Leader changed: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
```

Observe that:

- one node becomes leader
- other nodes act as followers

Observation: Leader election in 5-node cluster

Stop Two Nodes

Open another terminal
Run:
docker stop node4

docker stop node5

Check Running Containers After Stopping Two Nodes

Open another terminal
Run:
docker ps

Verify that only these containers are running:

- node1
- node2
- node3

Take Screenshot 14: Cluster running after stopping node4 and node5

```
C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>docker stop node4
node4

C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>docker stop node5
node5

C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>docker ps
CONTAINER ID   IMAGE                   COMMAND              CREATED         STATUS         PORTS     NAMES
d61948e3960f   raft_docker_lab-node1   "python node.py 1"   3 minutes ago   Up 3 minutes             node1
f7ece4e2cdd2   raft_docker_lab-node2   "python node.py 2"   3 minutes ago   Up 3 minutes             node2
3903be260355   raft_docker_lab-node3   "python node.py 3"   3 minutes ago   Up 3 minutes             node3
```

Observe Logs of Remaining Nodes

In separate terminals, run:

docker logs -f node1

docker logs -f node2

docker logs -f node3

Observe that the cluster is still active and one node is leader.

Cluster still working with 3 nodes

Stop One More Node

Open another terminal

Run:

docker stop node3

Check Running Containers After Breaking Majority

Open another terminal

Run:

docker ps

Verify that only these containers are running:

- node1
- node2

Take Screenshot 15: Only 2 nodes running after stopping node3

```
C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>docker stop node3
node3

C:\Users\SIRI DS\OneDrive\Desktop\raft_docker_lab>docker ps
CONTAINER ID   IMAGE                   COMMAND              CREATED         STATUS         PORTS     NAMES
d61948e3960f   raft_docker_lab-node1   "python node.py 1"   5 minutes ago   Up 5 minutes             node1
f7ece4e2cdd2   raft_docker_lab-node2   "python node.py 2"   5 minutes ago   Up 5 minutes             node2
```

Observe Logs After Majority is Broken

Open separate terminals

Run:

docker logs -f node1

Take Screenshot 16:

```
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[Node 1] Leader changed: None
[Node 1] No leader elected yet
[Node 1] No leader elected yet
[Node 1] No leader elected yet
[Node 1] No leader elected yet
[Node 1] No leader elected yet
[Node 1] No leader elected yet
```

Run:docker logs -f node2

Take Screenshot 17:

```
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[Node 2] Leader changed: None
[Node 2] No leader elected yet
[Node 2] No leader elected yet
[Node 2] No leader elected yet
[Node 2] No leader elected yet
[Node 2] No leader elected yet
[Node 2] No leader elected yet
[Node 2] No leader elected yet
```

Observe the output.

Logs after majority is broken

Stop the Cluster
Run:
docker compose down

19.Q&A

Why did leader election not occur when only 2 nodes were active in a 5-node cluster?

20. Result

The Raft distributed cluster was successfully implemented using Docker containers. The system performed leader election and demonstrated fault tolerance when the leader node failed. The experiment was extended to a 5-node cluster to observe quorum behavior. The system continued to function when two nodes were stopped, but became unavailable when three nodes were stopped, proving the majority requirement in Raft.

21. Conclusion

This experiment demonstrates how Docker can be used to simulate distributed systems and how the Raft consensus algorithm elects a leader and maintains system consistency. It also shows the importance of quorum (majority), where the system continues to operate only when the majority of nodes are available. When the majority is lost, the system becomes unavailable, ensuring consistency and preventing incorrect decisions.